PROGRAM GUIDE

EAI Robotics Future Founder Immersion Program

A 10-day journey where students build, pitch, and launch an AI robotics startup inside a real robotics company.

10 DAYS	REAL ROBOTS	REAL ENGINEERS	FINAL DEMO + PITCH

AGENTECH | FF

PROGRAM GUIDE

Program Overview

This guide provides a detailed 10-day structure for the EAI Robotics Future Founder Immersion Program. The program is designed for high school students who are ready to experience AI robotics as builders, product thinkers, and emerging founders.

The journey combines embodied AI, robotics engineering, mentorship, startup thinking, financial literacy, and a final hackathon. Students do not simply learn AI. They experience how an AI robotics venture is built from idea to prototype to demo to pitch.

Real Company Environment	Students enter a working robotics company context and learn from the culture, tools, and pace of real technical teams.
Founder-Level Journey	The program connects technical building with user problems, market logic, business models, and investor-style storytelling.
Final Demo and Pitch	The journey culminates in a hackathon, public demo, technical Q&A;, and founder-style presentation.

Daily Journey

Day 1: Welcome to the Future

Everything starts with curiosity.

Students enter the world of AI robotics through a real company environment. They meet mentors, explore the lab, form teams, and begin their journey from learner to builder.

Morning	Afternoon	Evening
- Opening Ceremony - Program Orientation - Meet the Mentors	- FF Lab & Facility Experience - Team Formation - Meet Your Robot	- Founder Fireside Chat

Learning Focus	Observe how a robotics company is organized, how teams communicate, and how engineering culture shapes product ambition.
Student Output	Program badge, team assignment, lab orientation notes, and a first reflection on a robotics startup idea.
Founder Lens	A founder begins by entering the world as it is, noticing real systems, real constraints, and real people.

Detailed Topic Pool

Day 1: Welcome to the Future

The topics below are optional areas the program may draw from. The final emphasis may vary based on student level, equipment availability, mentor focus, and project direction.

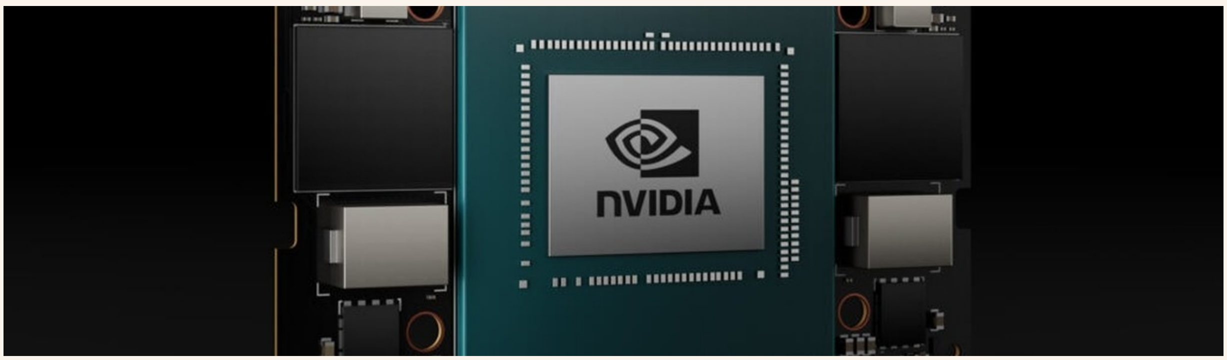
Session	Module	Possible topics
Morning	**Opening Ceremony**	- Program goals and culture - How the 10-day journey works - Lab safety expectations - Team collaboration norms - Final demo standards
Morning	**Program Orientation**	- Daily rhythm and milestones - Tool accounts and workspaces - Mentor feedback process - Portfolio expectations - How to document progress
Morning	**Meet the Mentors**	- AI mentor roles - Robotics mentor roles - Founder mentor roles - Office-hour etiquette - How to ask strong technical questions
Afternoon	**FF Lab & Facility Experience**	- Robotics lab stations - Perception and sensing demos - Hardware testing areas - Data collection workflow - How company labs reduce risk
Afternoon	**Team Formation**	- Strengths and interests map - Role selection - Team agreement - Decision-making process - Project theme matching
Afternoon	**Meet Your Robot**	- Robot anatomy - Sensor map - Actuator and joint overview - Battery and safety basics - Possible mission ideas
Evening	**Founder Fireside Chat**	- Founder origin story - Market timing - Building the first team - Raising early capital - Resilience through uncertainty

Daily Journey

Day 2: Think Like an AI Engineer

Learn how intelligent machines think.

Students learn the basic language of AI, robotics, and embodied intelligence. The goal is not only to understand concepts, but to begin thinking like an engineer.



Morning	Afternoon	Evening
- AI Fundamentals - Embodied AI Overview - Robot System Architecture	- Programming Workshop - Simulation Introduction - First Engineering Sprint	- Daily Demo & Reflection

Learning Focus	Connect AI concepts to perception, planning, control, and the practical limits of real robot systems.
Student Output	A first simulation workflow, a vocabulary map, and a team note on what makes embodied AI different from screen-based AI.
Founder Lens	Technical clarity becomes strategic clarity. Students learn to ask what the robot must sense, decide, and do.

Detailed Topic Pool

Day 2: Think Like an AI Engineer

The topics below are optional areas the program may draw from. The final emphasis may vary based on student level, equipment availability, mentor focus, and project direction.

Session	Module	Possible topics
Morning	**AI Fundamentals**	- Machine learning concepts - Neural networks - Embeddings and representation - Model evaluation - Failure modes and bias
Morning	**Embodied AI Overview**	- Perception-action loop - World models - Affordances - Planning and control - Sim-to-real transfer
Morning	**Robot System Architecture**	- Sensor layer - Compute layer - Middleware and APIs - Control loops - Telemetry and logs
Afternoon	**Programming Workshop**	- Python workflow - APIs and data structures - Version control basics - Debugging routines - Notebook versus script workflow
Afternoon	**Simulation Introduction**	- Digital twin concept - Scene setup - Robot model loading - Virtual sensors - Scenario testing
Afternoon	**First Engineering Sprint**	- Task breakdown - Issue tracking - Pair programming - Sprint demo - Blocker reporting
Evening	**Daily Demo & Reflection**	- Demo ritual - Learning log - Peer feedback - Risk list - Next-step planning

Daily Journey

Day 3: Build Your First Skill

Every robot begins with a single skill.

Students begin building reusable robot skills. They learn how perception, planning, and action connect together inside an embodied AI system.

Morning	Afternoon	Evening
- Computer Vision Basics - Sensors & Perception - Skill Graph Introduction	- Engineer-Led Skill Workshop - Build a Simple Robot Skill - Test in Simulation	- Guest Talk: Robotics in the Real World

Learning Focus	Translate sensor input into useful robot behavior and understand why reusable skills matter.
Student Output	A simple robot skill tested in simulation, with notes on failure cases and improvement paths.
Founder Lens	A product is built from small, repeatable capabilities that can be tested, improved, and explained.

Detailed Topic Pool

Day 3: Build Your First Skill

The topics below are optional areas the program may draw from. The final emphasis may vary based on student level, equipment availability, mentor focus, and project direction.

Session	Module	Possible topics
Morning	**Computer Vision Basics**	- Image pixels and channels - Edge detection - Object detection - Segmentation - Accuracy and confidence metrics
Morning	**Sensors & Perception**	- Camera input - Depth sensing - IMU basics - Lidar concepts - Sensor fusion
Morning	**Skill Graph Introduction**	- Skill nodes - Preconditions - Triggers - Action chains - Recovery paths
Afternoon	**Engineer-Led Skill Workshop**	- Read a sample skill - Modify parameters - Test robot behavior - Study failure cases - Mentor code review
Afternoon	**Build a Simple Robot Skill**	- Target selection - State machine design - Perception hook - Action command - Success condition
Afternoon	**Test in Simulation**	- Scenario variations - Log reading - Regression checks - Performance notes - Bug report writing
Evening	**Guest Talk: Robotics in the Real World**	- Deployment stories - Safety lessons - Customer needs - Product constraints - Robotics career paths

Daily Journey

Day 4: Connect AI to Reality

Ideas become reality when they move.

Students move from simulation into the physical world. They learn why real robots are harder than demos, and why engineering discipline matters.

Morning	Afternoon	Evening
- Hardware & Robot Integration - Safety and Testing Workflow - Real Robot Control Basics	- Deploy to Robot - Test, Debug, Improve - Engineering Review	- Lab Open Hour

Learning Focus	Experience the gap between simulation and real hardware, including safety, latency, calibration, and debugging.
Student Output	A deployment checklist, an engineering review log, and a revised prototype plan.
Founder Lens	Reality is the strongest investor. The system either works in the world, or it teaches the team what to fix.

Detailed Topic Pool

Day 4: Connect AI to Reality

The topics below are optional areas the program may draw from. The final emphasis may vary based on student level, equipment availability, mentor focus, and project direction.

Session	Module	Possible topics
Morning	**Hardware & Robot Integration**	- Actuator map - Sensor wiring - Compute modules - Latency budget - Power management
Morning	**Safety and Testing Workflow**	- Test zones - Emergency stop protocol - Checklist design - Fault isolation - Responsible escalation
Morning	**Real Robot Control Basics**	- Coordinate frames - Velocity commands - Motion limits - Teleoperation - Controller tuning
Afternoon	**Deploy to Robot**	- Package build - Configuration handoff - Network setup - Launch sequence - Rollback plan
Afternoon	**Test, Debug, Improve**	- Reproduce the bug - Inspect logs - Adjust parameters - Retest - Document results
Afternoon	**Engineering Review**	- Design rationale - Risk review - Mentor questions - Next actions - Acceptance criteria
Evening	**Lab Open Hour**	- Optional mentor help - Prototype cleanup - Experiment time - Team sync - Reflection notes

Daily Journey

Day 5: Design Like a Founder

Great engineers solve problems. Great founders find them.

Students learn that technology is only part of the company-building process. They explore customer needs, pricing, business models, and the basics of fundraising.

Morning	Afternoon	Evening
- Product Thinking - User Problems - Market Discovery	- Business Model Workshop - Financial Literacy for Startups - Startup Financing Basics	- VC / Founder Fireside Chat

Learning Focus	Move from a technical idea to a user problem, a market context, and a basic business model.
Student Output	A one-page problem statement, target user definition, early pricing hypothesis, and business model sketch.
Founder Lens	A strong founder does not start with a feature. A strong founder starts with a painful problem.

Detailed Topic Pool

Day 5: Design Like a Founder

The topics below are optional areas the program may draw from. The final emphasis may vary based on student level, equipment availability, mentor focus, and project direction.

Session	Module	Possible topics
Morning	**Product Thinking**	- Problem framing - User job - Product promise - Minimum viable product - Success metric
Morning	**User Problems**	- Interview design - Pain-point discovery - Use-case framing - Persona sketch - Evidence quality
Morning	**Market Discovery**	- Market map - Alternative solutions - Adoption barriers - Stakeholder roles - Competitor scan
Afternoon	**Business Model Workshop**	- Value chain - Pricing logic - Cost structure - Sales channel - Revenue assumptions
Afternoon	**Financial Literacy for Startups**	- Unit economics - Burn rate - Gross margin - Runway - Funding tradeoffs
Afternoon	**Startup Financing Basics**	- Angel versus VC - SAFE notes - Dilution - Milestone financing - Pitch materials
Evening	**VC / Founder Fireside Chat**	- Fundraising narrative - Investor questions - Timing - Team credibility - Ethical responsibility

Daily Journey

Day 6: Move Faster Together

Innovation is a team sport.

Teams accelerate their projects with engineer mentorship. Students learn how real teams plan, divide tasks, debug together, and communicate progress.

Morning	Afternoon	Evening
- Team Sprint Planning - Project Scope Review - Engineering Stand-up	- Project Development - Mentor Office Hours - Midpoint Technical Check	- Team Demo Practice

Learning Focus	Practice sprint planning, role division, engineering communication, and focused execution under constraints.
Student Output	A sprint board, midpoint demo, mentor feedback notes, and a refined project scope.
Founder Lens	Speed is not chaos. Speed comes from shared priorities, clear ownership, and fast feedback.

Detailed Topic Pool

Day 6: Move Faster Together

The topics below are optional areas the program may draw from. The final emphasis may vary based on student level, equipment availability, mentor focus, and project direction.

Session	Module	Possible topics
Morning	**Team Sprint Planning**	- Scope definition - Milestones - Role ownership - Bug triage - Daily goals
Morning	**Project Scope Review**	- Must-have versus nice-to-have - Technical feasibility - Demo boundary - Risk ranking - Mentor sign-off
Morning	**Engineering Stand-up**	- Yesterday, today, blockers - Dependency check - Decision log - Time-boxing - Team accountability
Afternoon	**Project Development**	- Feature implementation - Module integration - Testing - Demo interface - Documentation
Afternoon	**Mentor Office Hours**	- Technical choices - Architecture questions - Pitch logic - Failure unblocking - Next milestone planning
Afternoon	**Midpoint Technical Check**	- Prototype readiness - Sensor and data health - Integration risk - Demo quality - Pass-fail criteria
Evening	**Team Demo Practice**	- Timing - Narration - Live backup plan - Team handoff - Audience questions

Daily Journey

Day 7: Solve Real Problems

Technology matters when it solves something real.

Students connect their projects to real operational challenges. They refine use cases, test assumptions, and learn how robotics can create value in real environments.

Morning	Afternoon	Evening
- Real-World Robotics Challenges - Production Line / Workflow Perspective - Use Case Selection	- Project Deep Work - Technical Review - User Scenario Testing	- CTO / Expert Fireside Chat

Learning Focus	Understand robotics value through workflow, reliability, cost, safety, and operational fit.
Student Output	A selected use case, scenario test notes, and a value proposition statement.
Founder Lens	The best technical answer is not always the best business answer. The use case decides.

Detailed Topic Pool

Day 7: Solve Real Problems

The topics below are optional areas the program may draw from. The final emphasis may vary based on student level, equipment availability, mentor focus, and project direction.

Session	Module	Possible topics
Morning	**Real-World Robotics Challenges**	- Safety - Reliability - Edge cases - Maintenance - Environment variability
Morning	**Production Line / Workflow Perspective**	- Workflow mapping - Human-robot handoff - Throughput - Bottlenecks - Quality control
Morning	**Use Case Selection**	- Impact - Feasibility - Data access - User access - Pitch fit
Afternoon	**Project Deep Work**	- Implementation - Refactoring - Testing - Integration - Stabilization
Afternoon	**Technical Review**	- Architecture - Failure modes - Evaluation metrics - Mentor critique - Revision plan
Afternoon	**User Scenario Testing**	- Scenario script - Observation - Success criteria - Failure notes - Iteration plan
Evening	**CTO / Expert Fireside Chat**	- Technical roadmap - Hiring technical talent - Product architecture - Scaling systems - Responsible deployment

Daily Journey

Day 8: Polish Your Product

Details make products great.

Teams prepare for the final challenge. They improve their prototype, sharpen their story, and transform a technical project into a presentable product.

Morning	Afternoon	Evening
- Testing & Debugging - Product Polish - Demo Storyline	- Pitch Coaching - Presentation Design - Hackathon Preparation	- Hackathon Kickoff

Learning Focus	Turn a working prototype into a demo that can be understood by engineers, users, families, and guests.
Student Output	A polished demo flow, pitch outline, slide draft, and hackathon task list.
Founder Lens	A good demo is not a trick. It is a clear promise about what the product can become.

Detailed Topic Pool

Day 8: Polish Your Product

The topics below are optional areas the program may draw from. The final emphasis may vary based on student level, equipment availability, mentor focus, and project direction.

Session	Module	Possible topics
Morning	**Testing & Debugging**	- Test matrix - Log analysis - Root cause - Regression testing - Demo reliability
Morning	**Product Polish**	- Interface copy - Physical setup - Experience flow - Edge cases - Final checklist
Morning	**Demo Storyline**	- Hook - Problem - Solution - Proof - Closing ask
Afternoon	**Pitch Coaching**	- Audience lens - Product narrative - Objection handling - Traction proxy - Founder presence
Afternoon	**Presentation Design**	- Slide hierarchy - Visual proof - Concise copy - Data chart - Rehearsal flow
Afternoon	**Hackathon Preparation**	- Scope lock - Team roles - Asset list - Risk plan - Time-boxing
Evening	**Hackathon Kickoff**	- Rules - Judging criteria - Timeline - Team commitments - First sprint

Daily Journey

Day 9: Build Under Pressure

Build fast. Learn faster.

The two-day hackathon begins. Teams work under pressure, make tradeoffs, solve unexpected problems, and prepare their final demo.

Morning	Afternoon	Evening
- Hackathon Sprint - Mentor Check-ins - Rapid Prototyping	- Hackathon Development - Testing & Debugging - Final Submission Prep	- Final Build Lock - Demo Rehearsal

Learning Focus	Make fast decisions, stabilize the build, and separate must-have functionality from nice-to-have ideas.
Student Output	A final prototype submission, demo rehearsal notes, and a last risk list.
Founder Lens	Pressure reveals priorities. Teams learn what to cut, what to protect, and what to explain clearly.

Detailed Topic Pool

Day 9: Build Under Pressure

The topics below are optional areas the program may draw from. The final emphasis may vary based on student level, equipment availability, mentor focus, and project direction.

Session	Module	Possible topics
Morning	**Hackathon Sprint**	- Rapid planning - Execution blocks - Mentor checkpoints - Daily target - Risk cut
Morning	**Mentor Check-ins**	- Short update - Problem escalation - Decision support - Technical feedback - Pitch feedback
Morning	**Rapid Prototyping**	- Paper prototype - Code stub - Sensor mock - Demo harness - Fast testing
Afternoon	**Hackathon Development**	- Integration - Interface - Data pipeline - Robot behavior - Slide sync
Afternoon	**Testing & Debugging**	- Test matrix - Log analysis - Root cause - Regression testing - Demo reliability

Detailed Topic Pool Continued

Day 9: Build Under Pressure

The topics below are optional areas the program may draw from. The final emphasis may vary based on student level, equipment availability, mentor focus, and project direction.

Session	Module	Possible topics
Afternoon	**Final Submission Prep**	- Asset packaging - README - Backup demo video - Run sheet - Submission checklist
Evening	**Final Build Lock**	- Freeze rules - Smoke test - Fallback mode - Version tag - Equipment check
Evening	**Demo Rehearsal**	- Timing - Speaker roles - Q&A; practice - Transitions - Confidence

Daily Journey

Day 10: Launch Your Future

Your journey has just begun.

Students present their work to mentors, engineers, guests, and families. The program ends not as a graduation, but as the beginning of a longer path into AI, robotics, research, entrepreneurship, and real-world impact.

Morning	Afternoon	Evening
- Final Demo - Product Presentation - Technical Q&A;	- Investor-Style Pitch - Awards - Closing Ceremony	- Networking - Certificate - Alumni Invitation

Learning Focus	Present technical work with confidence, answer questions, and connect the project to a future pathway.
Student Output	Final demo, founder-style pitch, certificate, portfolio material, and alumni pathway invitation.
Founder Lens	Launch is not the end. It is the first public signal that the team can build, learn, and continue.

Detailed Topic Pool

Day 10: Launch Your Future

The topics below are optional areas the program may draw from. The final emphasis may vary based on student level, equipment availability, mentor focus, and project direction.

Session	Module	Possible topics
Morning	**Final Demo**	- Live robot run - Recorded backup - Use case framing - Technical walk-through - Value story
Morning	**Product Presentation**	- Problem statement - Target user - Demo proof - Business logic - Roadmap
Morning	**Technical Q&A;**	- Architecture - Data - Testing - Failure analysis - Safety
Afternoon	**Investor-Style Pitch**	- Market - Moat - Traction proxy - Business model - Funding story
Afternoon	**Awards**	- Judging categories - Mentor feedback - Recognition - Next path - Celebration

Detailed Topic Pool Continued

Day 10: Launch Your Future

The topics below are optional areas the program may draw from. The final emphasis may vary based on student level, equipment availability, mentor focus, and project direction.

Session	Module	Possible topics
Afternoon	**Closing Ceremony**	- Reflection - Certificates - Family photos - Mentor thanks - Alumni invitation
Evening	**Networking**	- Mentor conversations - Engineer conversations - Family showcase - Contact etiquette - Follow-up message
Evening	**Certificate**	- Program record - Project title - Skill badges - Portfolio language - Next steps
Evening	**Alumni Invitation**	- AI Club pathway - Internship pathway - Research pathway - Future events - Mentorship

PROGRAM GUIDE

Student Outcomes and Next Pathways

By the end of the program, students should leave with tangible work, stronger technical confidence, and a clearer sense of how AI robotics connects to research, entrepreneurship, and real-world impact.

01	A completed AI robotics project
02	Final demo presentation
03	Founder-style pitch experience
04	Certificate of completion
05	Project portfolio material
06	Exposure to real engineering workflow
07	Invitation to continue through Agentech AI Club / Internship / Research Pathway

Internal program guide for families, students, mentors, and partners.